

ActInf Livestream #052.2 ~ Geometric Methods for Sampling, Optimisation

Livestream | Geometric Methods for Sampling, Optimisation, Inference, Adaptive... | 1:56:25

Hello and welcome. It's ACTIMF livestream 52.2 on March 9th, 2023. Welcome to the ACTIMF Institute.

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applied active inference. You can find more information on these links. This is a recorded

and archived live stream so please provide feedback so we can improve our work. All backgrounds and

perspectives are welcome and we'll be following video etiquette for live streams. Head over to

activeinference.org to learn more about getting involved with projects and learning groups.

All right, we are back in livestream 52.2. In our third discussion on the paper geometric methods

for sampling, optimization, inference, and adaptive agents, we had a .0 background and context video

and last week Lance joined for 52.1 where we had a great overview discussion on the paper

so today we're going to see where it goes, see where our last week has taken us and how we're

thinking about it or curious about it. If you're watching live, of course, feel free to write

questions in a live chat. Otherwise, let us pick up on a mostly blank slide and uh just thanks again

Lance for joining. If you want to give any sort of introduction or recap opening here, go for it.

Lance

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Male

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principal literature and active inference are used to because here the goal of inference is about approximating expectations as opposed to just approximating distributions in of themselves it turns out to be these two perspectives turn out to be dual but i guess here we want to develop notions of divergences and discrepancies that are a bit more general than the kl divergence and that can use to solve problems that the kl divergence cannot and i guess the overall picture for what we want to do that is the kill divergence turns out to have a lot of really nice properties that we can discuss one of them is that if the kill divergence is reduced it means that that the two distributions in play are more similar in terms of information so there is this idea of information monotonic monotonicity where the kl divergence sort of yeah given ordering as to what extent two distributions quantify i mean so if you have three distributions and you compute the kl divergence between them and now kl of a b is lower than kl of ac it means that b is more closer in terms of information to a than c is to a so you have this really nice thing that is captured by the kl divergence with which makes sense when we're dealing with information the kl divergence also has so many other nice properties um it it's not a distance but it turns out to behave a bit like a square distance so you have this kind of like really nice pythagorean theorem um and i won't get into the exact statement but it's like if you have uh a b and c distributions then $kl(a, b) + kl(a, c) = kl(b, c)$ if you have like a rectangle triangle if abc define a rectangle triangle in information space um you have other properties like the kl divergence gives and so on um so the kl divergence is in general i would say the distribution the divergence of choice but it turns out that in many cases you just can't use it for example when you have samples and you want to approximate some samples with a distribution then the kl leverage just is not going to worry there so you need to derive some other things so this is to say that we're considering statistical inference a bit more generally than what we do in active inference in general and so this speaks to why the uh the section here is a bit different from the standard uh inference literature that that we usually consider and then i think that the well then this section 5 which is about active inference and i think we should discuss that a little bit more because the formation of active inference that's presented there is to my mind the most general and also the simplest concept conceptually that's been out there i mean we sort of like recap the derivation of active inference and also like the properties of the expected energy properties of active inference and also how to scale active inference and so on and we do that in just seven pages or five or five pages i mean it's just very short um so it's dense sure it's like a really concise summary and actually from there you can really derive a lot of you know uh technical papers on active inference it's like to me it's the most general and if if you as a reader can understand that section then you can sort of really understand what active inference is about um so yeah this is kind of the overview for today um i would really like to be discussing questions because last time we really discussed about most of the papers so but yeah whatever comes up awesome all right great well let us uh talk about some of the more foundational pieces of the points that you raised about the kl which generalized inference beyond how it may have been brought up in other act inf and then we can spend most of the time in section five and looking at some of those figures connecting some of the intuitions about the ball rolling down the bowl to the person running

so sounds good on hamiltonian monte carlo where do you want to pick up sure i mean so i mean donian so last time we discussed a lot about the problem of sampling and why that's a difficult problem and we arrived at the conclusion or or i presented like monte carlo methods how they work so you're basically running um a stochastic process like a random random motion and sort of the this distribution defining the process is going to convert to a target distribution which means that when you run the process long enough then every point that it's going to be in is going to be like a sample of the distribution that you want to sample from um now there's a lot of issues with that um i mean conceptually it's not so difficult but actually when you want to implement this in practice it turns out to be really hard because if you just implement the simplest stochastic process to sample your target distribution um it's going to be extremely slow and i think that's the main bottleneck when developing monte carlo methods monte carlo sampling in in general is slow and that's also one of the reason one of the reasons why one might want to do variational inference instead of sampling um so something is slower but it's also more accurate it can approximate our distributions that are completely arbitrary so if you care about accuracy then and you have time and computational resources then for sure go for sampling if you care about speed about doing things online and you don't care about accuracy so much then the right no inference is the way to go at least that's my understanding right now so um let's say you wanted to sample a distribution in a continuous space so it could be um just as last time let's imagine the state space is the desk where where i'm at and you sort of had this distribution which could be like multimodal uh very weird and you just want to take samples from there um hamiltonia monte carlo is probably the state of the art methods method to do that there's a lot of other methods out there but hamiltonia monte carlo typically just works very well um and in a wide variety of situations and so the idea of hamiltonia monte carlo is you're gonna augment the state space with um so let's let's say that the original state space you start with are the positions and so you're gonna augment that with a velocity state space so you kind of if your original state space was euclidean space of n dimensions you just end up with a nuclear space of two n dimensions so you double the size of the state space and now you say okay well um the distribution that i want to sample from actually defines an energy landscape um so it technically technically it's like if you take if you have distribution which is p then $-\log p$ is an energy landscape so um points where $-\log p$ is low are points where p is high and so these are points that you want to sample a lot from and contrary wise um if $-\log p$ is very high then p is low and you don't really want to go there so much because these are points of low probability so i think this $-\log p$ actually there's many reasons why $-\log p$ is is meaningful in physics but just note for now that $-\log p$ is just like a function and the minimum of the function of the of that function at a high probability point and uh these are high these are high energy points where you basically don't want to go to go there so much um so let's call this $-\log p$ potential energy and so this is what this really is in physics if p technically in physics p would be gives measure um what people call an equilibrium distribution um last time we saw like soft max of minus something the something is the is the potential energy um anyway and so $-\log p$ would

be the potential energy and then if you add kinetic energy on your velocities u then you get what's called the hamiltonian which is the sum of the potential and kinetic energies so what is exactly the kinetic energy the kinetic energy is like velocity squared pretty much so if you add remember that your state space is position and velocity if you add and you take a point in state space the uh so you would have u kinetic energy which is velocity squared and a potential energy which would be minus log t now if you add the two together what you get is a hamiltonian which is the sum of the kinetic energy and potential energy and this is just standard physics hamiltonian is a sum of kinetic and potential energy and it gives us the total energy of the system u so why we started with the problem of sampling and here i just told you something very complicated where we get a hamiltonian there's actually a good reason why we want to do that and it comes back to the idea of geometric integration that we talked about last time which is that u typically maybe i have a process that i know would will give very efficient sampling but actually when i implement it on the computer in discrete time i just lose all the all the properties that make that something efficient so actually it turns out that most people working in monte carlo sampling they're working on efficient discretizations of continuous processes as opposed to on the continuous processes themselves really the bottleneck and the difficulty here is is the implementation part uh implementing process on on a computer like you and i had in such a way that we retain all the good sampling properties u so one very powerful idea is that of geometric integration which is of preserving geometric properties of a system i mean geometric integration is the field that develops u computational methods of uh numerical integration and numerical discretization in such a way that they preserve important geometric properties of the system here in the gym the geometry in play is the hamiltonian and so you might think well that's pretty weird i mean a hamiltonian is an energy and here we're talking about preserving the geometry and preserving the hamiltonian how do these two things fit together and so it turns out that the presence of a hamiltonian and the fact that we have a state space that has positions and velocity technically in mathematics what what we then get is what people call symplectic geometry so symplectic geometry is just arises when we have those state space that comprise positions and velocity and when where you have hamiltonians so this is just like a way of explaining why the hamiltonian uh why the hamiltonian is like closely and intimately related to geometry so geometric integration enables you to u to yeah to discretize processes in such a way that the hamiltonian is preserved now when you think about the hamiltonian the hamiltonian gives you the energy of a particular point in state space and so if you simulate trajectories that preserve the hamiltonian what you're effectively doing is sampling from the contours of the probability distribution contours that have the same probability so you're basically going around let's say for example in circles u in a region that has the same probability when we want to sample we want to go everywhere i mean we want to sample regions of high probability low probability and want to be able to go from one to the other so what geometric integration allows us to do is to simulate dynamics that are going to preserve the contours of the probability distribution and they're going to do so very well the advantage here is that

when we use geometric integration to simulate hamiltonian dynamics which are conservative and again stay in the contours um what geometric integration allows us to do is to take time steps that are very long so it enables us to travel very far in the landscape while still preserving the hamiltonian so you don't need to take very small time steps to remain accurate and preserve the hamiltonian but actually geometric integration allows you to you know take very long time steps and still preserve the hamiltonian so now we have a dynamic that's hamiltonian preserving that's just very good because you can take very long time steps and that enables you to go around the contours of the probability distribution so what you want to do next is you know to change contours you want to go to contours that are of higher probability contours of lower probability and you want to do so in a way that the contours of high probability are sampled much more and much more often than the contours of lower probability so what you do is that you augment your hamiltonian dynamic with what people call a velocity refreshment or a momentum refreshment and so this is how it works how it works um so you're gonna yeah simulate your hamiltonian dynamic for for some time and then after after a while you're gonna be like okay well now i'm gonna i'm gonna change the velocity at random by by just drawing a new velocity from a gaussian distribution and so by drawing a new velocity from a gaussian distribution you're just gonna change contour completely and then there's what's called um and then there's the constraint that i just talked about which is you want to change contours of the probability distribution in such a way that contours of high probability are visited much more often than contours of low probability so you want to preserve that you want to do that um proportionately in a way so the third ingredient of hamiltonia monte carlo comes in which is uh metropolis hastings and so metropolis hastings is this accept rejects that that we discussed briefly last time and so metropolis hasting is just super clever and it enables you to to say um um to it it just tells you whether this um this momentum refreshing step is actually a good one or a bad one and should be rejected a momentum refreshment is good if overall your sampling is gonna remain um is gonna remain faithful to the target distribution but it is bad if it remains unfaithful useful so actually um running this metropolis hastings except rejects that allows you to change contours of the probability distribution in a way that remains faithful to the target distribution and so there you go that's basically hamiltonia monte carlo so to recap you augment the state space by adding velocities uh this allows you to build a hamiltonian by declaring that the original probability distribution gives you potential energy and you add a kinetic energy on the on the velocities which is just velocity squared because you have a hamiltonian you can um you can use geometric integration to simulate hamiltonian dynamics very accurately and with very long time steps the very long time steps is a crucial thing because it means that with very low computational cost you can sample very far so it means that you don't get stuck in a region of the probability distribution but you can actually visit it much more fast and efficiently so that's the first part um the problem with those dynamics again is that they remain on the contour of the probability distribution and so you want to sample the whole probability

distribution so what you do is every once in a while and i think that's a hyper parameter in your in your sampling algorithm let's say every 10 iterations of the hamiltonian dynamic every 10 time steps of the hamiltonian dynamic you're going to sample um you're just going to randomly take a new velocity so you you're going to change contouring the probability distribution and the crucial point there is you want to change contour in a way that's faithful of the probability distribution and that's faithful of the sampling problem that you want to do there comes the last thing which is metropolis hastings which ensures that the momentum refreshment will be uh good for sampling i mean will preserve your target probability distribution so with that you just get a very efficient um sampling algorithm and so it seems a bit convoluted right um the and one bottleneck of course is that you need to double the dimension of the state space and if your state space is already extremely large that could be a bottleneck um it could be the case that actually doubling the dimension to add velocities uh it could be a computational bottleneck so that's

a problem um but still in most cases that's not a problem and um so again hamiltonian monte carlo is convoluted but the overall take-home message i i would want you to take from this is that it is a method that

that remains faithful to the probability distribution that i want to sample and this is crucial and it is also a method that through geometric integration it enables you to take time steps that are very long and still get accurate sampling so this is the the really cool thing um if you you could come up with a whole bunch of other methods that did not require to you know double the dimension of the state space or that didn't use geometric integration the problem that you would probably run into is that the over that that the thing that you came up with when you implement it in the computer it doesn't exactly preserve the the the probability distribution that you want to sample and so the problem with that is that this would lead to biased sampling and by a sampling by something is like you're going to you're going to you're going to you're going to sample a different probability distribution which might be just a bit different but still um a different probability distribution than what you really want to sample and this would bias your your predictions so the really cool thing of hamiltonian monte carlo is that you're actually able to have unbiased sampling through the metropolis hastings step

um and this is the crucial thing so it's computationally implementable in general there's no computational bottlenecks apart from this doubling of state space and it leads to unbiased sampling and it's also relatively simple so this is why this is really used all over the place there's um there's another few perspectives that might explain why it works so well so in in the paper in in section in the last subsection of of uh

of section two which is about optimization um so the the section on optimization is about how to accelerate optimization

how to derive an optimization algorithm um yeah yeah 2.6 yeah that's the one so the whole thing about section two

is about deriving an optimization algorithm that accelerated in a physical way and by mean accelerate what i mean by accelerated it's not about just going faster but it's about having acceleration

um which means that um so if you go slightly up in the paper on the figure i think this figure is great and by the way i wasn't the one who who came up with this figure but i think i think this figure is great it really gives a lot of intuition for what acceleration really is so if you if you look at the at the green cup i don't know how you call this at the green well on the left uh this would be like um a ball rolling down uh the well in like uh and the well is filled up honey to to like a certain level and so there you get a lot of friction so your ball would just roll down very very slowly and the speed of the ball would be proportionate to the slope of the well um so this is not accelerated because there's a lot of friction and so the speed is just proportional to the slope of the well and this so this is actually what grade in the sand does but now if you go on the right you get what people call in physics on underdamped system which is also an accelerated system a system that's meaningfully accelerated so if you replace the honey in the well by some water then there's going to be way less friction and your ball is actually going to accelerate and overshoot the minimum of the well and then sort of stabilize but the point here is that this um this bowl is just going to get so much faster to the minimum now um so let's say that we just um i mean so this optimization this idea for optimization is is just extremely powerful and um and by the way on the right hand side in the graph you can see the improvements that you get when you when you implement these sort of ideas so this was on um i'll come back to to sampling in a bit i promise but this is actually very well very related you you can see on the graph on the right what kind of improvements you get when you go from a under that first order system grading the sand system to a what when you go from an overdamped first order building the sun system to an underdamped second order accelerated system and so you can see the curve right um so the black slash orange curve is overdone there's no acceleration it's very slow um the the last curve the the blue one it's accelerated and it's just super fast so you can see it sort of gives you like um quantitative uh data as to how many improvements you can get by implementing this um acceleration in a physically meaningful way could you describe the axes of the graph and also what the lee group is here um so the lee group is um so in in the figure legend in the fourth line you see the lee group is so n so that's this um technically that's the special or orthogonal group of dimensions n and so if i if i remember correctly these are all the matrices all the square matrices of size n by n that have a determinant equal to one so the determinant is just like um a function where you give it a matrix and it gives you a number the fact that um the fact that the determinant is equal to one means that here we're just taking matrices which are um which produce trend not translations but um changes of coordinates that preserve the geometry in a system so so so just to to elaborate on this a little bit because for people who are not intimately familiar with matrices it might seem a little opaque so when you have a matrix a matrix can multiply vectors so a square matrix takes in the vectors of size of dimension n and it outputs vectors of dimension n by multiplication so if you have a square matrix matrix it just induces a transformation of your state space because each each vector each point which is a vector gets converted into another point which is another vector now um the so there's a lot of i guess properties there right you could have all sorts of translations of state space or rotations in state space or transformations of state space if the determinant of the matrix is equal to one it means that the

transformation of state space that the matrix induces U will preserve the geometry of the six places it will preserve distances if you and it will preserve crucially orientation as well because if the if you only asked $\det U$ to be equal to one or minus one then you would the the matrix transformation would preserve the geometry but it would not preserve the orientation it could like mirror things $\det U$ so if you only asked the determinant to be equal to one or minus one you would get a lee group that's called the orthogonal group which is usually denoted $O(n)$ and here we we require the the determinant to be just equal to one and so you get the special orthogonal group and so it's this group of matrices that have this property

now this is not $\det U$ really i mean this is not really important for the example here because you could have taken any other d group and got something similar

U and got a similar difference in performance but but $\det U$ yeah i guess it's just interesting for its own take

$\det U$ and so the graph though the k is the number of iterations so the number of time steps so that's the x-axis and the $\det U$ the y-axis is how close you are from the minimum

$\det U$ and and it's in the log scale so as you can see the the blue curve is going to get like in 100 iterations

it's going to get basically a 10^{-4} 10^{-5} distance from the minimum so it's like

super super close it's basically gets there in 100 iterations U when you take the other normal gradient

descent methods where you see that it's going to take so much longer $\det U$ so yeah this is really the

advantage of using second order methods and what i mean by second order methods is that by second order

means that you actually double the state space to introduce velocities so that you not only have a

dynamical system over positions but you have a dynamical system of velocities and so it means that you have a

a physically $\det U$ a physical notion of acceleration because acceleration is a movement of velocities

$\det U$ and so if you double the state space by saying this is position this is velocities and you say i have

a motion in this like a state space that's twice as big then you have a meaningfully a meaningful notion

of acceleration and this is really powerful and so you can see already a parallel here which is

that Hamiltonian Monte Carlo also has this notion of acceleration in some way at least just intuitively

because we we also U double the size of the state space to introduce velocities and so it turns out

that this intuition is actually U you can make it into a formal correspondence $\det U$ and this is i think this

is something that quite interests me to be honest $\det U$ so if you if you remember if we come back to the

intuition for sampling the intuition for something through Monte Carlo methods is we have a target

distribution that we want to sample from so we're going to run a dynamic a random dynamic towards that

towards that distribution now what makes sampling efficient is that the distribution that characterizes the

process because again the process is random so each time it is at a random location that is characterized

by a distribution what makes sampling efficient is that the distribution characterizing the process just

converges as fast as possible to the thing that you want to sample from so there is this is to say that

you can think of sampling as an optimization on the space of probability distributions you want to

move your probability distribution of the process as fast as possible to your target and so you can see

from there that sampling is actually not so different from variational inference $\det U$ it's just the same idea

only variational inference you're typically gonna like $\det U$ yeah take a parameterized family of distributions and

approximate the target while here you have a non-parameterized family that's given by your dynamic that's going

to perfectly match the target μ so so again so so with that i want you to have the take-home message that sampling you can think about it as an optimization on the space of probability distributions in the sense that you have a target distribution and you want to get there as fast as you can with the process now

let's suppose and so here is the crucial connection between sampling and optimization in in the way that we've

described it here

let's suppose that you run this accelerated optimization scheme on the space of probability distributions

then the the process so you would get the dynamic on the space of the probability distributions that has a meaningful notion of acceleration μ through through the accelerated method that that was shown here shown here so if you look at what that what kind of dynamic this really gives you it gives you a process which is given by a stochastic differential equation that is known as underdamped langevin dynamics μ so there's an equation for it in i think section subsection 2.8 but the point is underdamped langevin dynamics is a stochastic differential equation whose density or probability distribution

stoles this accelerated optimization problem on the space of probability distributions

so just from there you know that underdamped langevin dynamics is going to be a very efficient sampler because it meaningfully accelerates and gets to the target i mean quite fast the problem is underdamped langevin dynamics you cannot simulate it accurately well you can simulate it accurately but you cannot simulate

it exactly on your computer in practice so this comes back to the numerical integration problem that that we discussed just before μ and so you have to to find a way to to discretize or or in other words implement underdamped langevin dynamics on your computer in such a way that you retain that the thing that you implemented on the computer retains this meaningful notion of acceleration and so it turns out that you can see on Hamiltonian Monte Carlo as a faithful numerical discretization or numerical implementation of underdamped dynamics that's going to preserve these acceleration properties and therefore

these this efficient sampling so this is all like μ yeah a lot of interesting connections but basically what what i want to get at is you really have this notion of acceleration that permeates well i guess physics but here it permeates sampling and it permeates optimization and so the method that was shown here about optimization and Hamiltonian Monte Carlo they're just the same in a way only one is applied to optimization the other is applied to optimization on probability distributions of aka sampling the other is applied to a higher level and so the other is applied to a higher level and so the shadow Hamiltonian is what is the shadow Hamiltonian and why does it sound so cool

i know right yeah it's uh it's really cool μ it's a recall name so the shadow Hamiltonian is μ okay so so let's go back to the Hamiltonian μ so we have a Hamiltonian and we want to simulate the dynamic that preserve the Hamiltonian we want to implement that on the computer we're going to do that through

geometric integration geometric integration gives you a bunch of algorithms to preserve the Hamiltonian i mean a bunch of algorithms they can implement on the computer and they're going to preserve the Hamiltonian um do they actually really preserve the Hamiltonian it turns out that no so what i what i said before is a bit of a shortcut like um because the numerical integration or that you get through any any numerical method including geometric integration is not going to exactly preserve the Hamiltonian but it's going to preserve what people call a shadow Hamiltonian which is almost the same as the Hamiltonian but with extra terms that sort of vanish if the if the time step is very is very short um so this is to say that your numerical dynamic uh implemented through geometric integration is going to exactly preserve the shadow Hamiltonian and approximately preserve the true Hamiltonian and so there's um in the papers we show um i don't know if it's well it's probably algorithm independent um but basically depending on the algorithm that you choose you you want to show to what extent the shadow Hamiltonian truly approximates the true Hamiltonian the the virtue of geometric integration methods is that actually the shadow Hamiltonian turns out to be extremely close to true Hamiltonian which means that you can take very long time steps and still be very good at approximately at preserving the true Hamiltonian so here you don't when you implement these methods you don't exactly preserve the true Hamiltonian um but you do you still do it pretty well and so in Hamiltonian Monte Carlo the momentum refreshment and metropolis usually the metropolis hastings actually accept rejects that step is going to correct it is going to correct for those failures of of truly preserving the Hamiltonian so even though you have some you don't you don't exactly preserve the Hamiltonian in your numerical integration in Hamiltonian Monte Carlo the metropolis hastings accept reject step is going to correct for this inaccuracy. And so this is really the true beauty of Hamiltonian Monte Carlo is that even though you get a lot of things that are not exactly preserved when you implement things on a computer, thanks to Metropolis Hastings, overall your sampling is going to be perfect in the sense that you're going to truly preserve your target distribution. So this is really the key. Now a follow-up question to that is, okay, well, if Metropolis Hastings is just so crucial and it just gives your dynamic the property that it's going to preserve whatever it does, even if it samples really badly, if you add Metropolis Hastings, it's still going to preserve your target distribution, then why can't you just come up with an average process, like whatever kind of process, and add Metropolis Hastings at the end. And so you can do that, you can take any kind of process. And after and to let's say you take a random process, it could be the worst in the world, let's say you wanted to sample from a Gaussian, and you actually take a Brownian motion. Now this is clearly not going to work because Brownian motion just goes all over the place, it could go infinitely far Brownian motion, and it's just like random motion, right, completely random motion. No, there is structure to it, but there's no, the structure in Brownian motion means that it's such that Brownian motion is just going to just spread really, really far. So Brownian motion is not going to preserve your target distribution. So it's going to, and it's also going to be very slow, by the way. So it's not going to be a good sampler. If you add Metropolis Hastings on Brownian motion, which is you integrate Brownian motion. So you

simulate a step of Brownian motion on the computer, and then you do Metropolis Hastings to use to know whether you should accept or reject that step. Metropolis Hastings will tell you whether you should do it or not, you either accept and you stay and you keep going from that new position or you reject and you start from where you started from and take a new sample, accept reject. If you accept, you keep going. If you reject, you go back and you sort of go like that. So if you do Metropolis Hastings there, your stumbler is going to be exact. So it is going to preserve the target. And so you're going to have accurate sampling. And so this is crucial. This really highlights the importance of Metropolis Hastings. But there is a big but your your stumbler here is going to be extremely slow. And it's going to be extremely slow, first of all, because Brownian motion in and of itself, it doesn't at all preserve the target distribution. I mean, just goes all over the place. And you just want to sample a lot around the mode of the Gaussian, for example. So it means that Metropolis Hastings will reject a lot of your steps, because a little of your steps will try to go further away while you want to stay around the mode, typically of the Gaussian. So you're going to do a lot of steps for nothing.

So you're going to do a lot of steps for the sample. And also, Brownian motion just does not have the qualities that make a sampler efficient. And so the reason why Hamiltonian Monte Carlo is so good is because it's able to integrate Metropolis Hastings and still preserve a lot of other properties that make that make the sampler efficient. So if we remember Hamiltonian Monte Carlo has this geometric integration that does not exactly preserve the Hamiltonian, but it does still does it pretty well approximately by preserving a shadow Hamiltonian. So this means already that that integration step is going to be really good. So there's going to be a lot of acceptance in the Metropolis Hastings.

So and also the way the way the scheme is set up, there's going to be a lot of acceptance step in the Metropolis Hastings. So it means that most of the samples that you would draw will actually be used instead of being rejected and you have to start over again. So that's one advantage. The other the other advantage of Hamiltonian Monte Carlo and this is a crucial one and we kind of discussed this last week is that one crucial thing to be a good sampler is this idea of time irreversibility. And so I'll emphasize it again because it's really crucial and there's a lot of literature on this and it's something that we reviewed also in the paper. So a sampler that is time reversible even if that is time irreversible. So what do we mean by this? A sample is time irreversible. If and only if were you to play the dynamics forward or backward there will be yeah so yeah so so going on your bullet point here a sample is time reversible is time reversible if you were to play the dynamics forward or backward in time so forward in time and then maybe you would play them by reversing the movie by playing the movie backward

if you were to do that then you then the the two movies that you would see would be qualitatively the same and statistically the same. So a process is time reversible if and only if the process if you're running forward or backward it's basically statistically the same. Now what does this mean in practice? Well if your process is time reversible then it's going to backtrack very often actually

So it means that you you would be somewhere in the probability distribution just sampling there and you will go forward and then you probably go backward and so on and you kind of get stuck in a region

until you go somewhere else but it's just going to be very slow to move around and so it's going to be very slow to get you know good sampling because it's going to take you a long time to visit all the regions of the probe into distribution. In other words the distribution characterizing the process it's not it's going to move very slowly to the target distribution that's another way to look at it. If the process is time irreversible on the other hand then it's a lot less likely to go backward during the sampling process so it means that it's going to visit the target the target distribution a lot more I mean it's just going to go around imagine if you're not allowed to go backward just as a human being and you're walking around you're just going to end up in many more places than if you were allowed to go backward and if you were and if you were just going backward all the time to where you started then you wouldn't be able to do a lot of visiting so it would be a bad sampler so so I guess there's some like straightforward intuition there so one idea is in sampling is we want to optimize the extent to which a sampler is time irreversible we want to increase time irreversibility as much as we can to force the sampler to just move around as much as possible so that's an idea and and it is explored currently I guess it's a bit of an open problem of how do you really do that and how do you implement that on a computer in a way that you know works and preserve all the properties but the point I want to I wanted to get to is that metropolis hastings is a blessing and it's also a curse it is a blessing because when you add it to any kind of process you will make your sampling unbiased so it means that you will sample the right distribution even though you're implementing this on a computer and there can be a lot so many issues with the numerical numerical integration numerical approximation and so on but it's also a curse because whenever you've had metropolis hastings um on on a process it's going to make it time reversible so actually yeah if you if you just took a random process and added metropolis hastings you would get something that's inevitably going to be quite slow and so now you get to the you know the problem or the conundrum where what what do I do do I go for unbiased sampling do I care about accuracy and then I should add a metropolis hastings or do I not care about accuracy so much and then I should not have metropolis space hastings and it's going to go faster generally and so Hamiltonian Monte Carlo actually has the blessing and does not have the curse and this is why Hamiltonian Monte Carlo is just so good and again it's also complementary to all of the things that we've been discussing and the reason why Hamiltonian Monte Carlo is blessed and not cursed is because the momentum the how do you call this the metropolis hastings is just done on the momentum refreshment step as opposed to being done on the overall dynamic so because if you remember you do geometric integration to simulate Hamiltonian dynamics now every once in a while you will take a momentum refreshment and then do metropolis hastings on that momentum refreshment as opposed to doing metropolis hastings on both the Hamiltonian dynamic and the momentum refreshment and so it's not it's not something that I can really explain how how that works but it turns out that by just doing the metropolis I mean Hamiltonian Monte Carlo is a way of having metropolis hastings in a way that you preserve uh the unbiasedness that's so that your sampling is accurate but you don't sacrifice the time irreversibility

and so and so you get both um and this is because the metropolis hastings is just on one of the components and not on both

so to kind of connect that to an example as we move into active inference

here we have some probability isocontours

some we have the high probability carpool lane and then we have some lower probability lanes

and uh we can go different speeds in these lanes and we want to have a full accelerated model here

it's almost like the mh is refreshing our velocity just asking when we want to change the lanes but

it's not our full self-driving car metropolis hastings algorithm but we're able to use our position and

acceleration when we're in our lane to take full advantage of the acceleration following the shadow

road not necessarily the true road but the shadow road is close enough or the shadows on the road and then

lane changes are these computationally costly and reversible but still super useful propositions

exactly and they actually they're actually not computational league costly they would be if you

were to reject many samples uh but here you in Hamiltonian Monte Carlo typically you don't get to

reject many samples so yeah you that's such a i mean that's such a great picture so the car follows the

shadow road by just doing geometric integration and following the shadow Hamiltonian every once in a

a while you get a momentum refreshment a velocity refreshment which is oh um now i move to the

first lane or the second lane third lane and so on and you get a metropolis hastings correction step that

tells that says do i accept this proposition proposition proposal of lane change or do i reject it and so on it goes

so how does this connect to active inference

well i mean great question uh so inactive inference inactive inference proper you don't have any sampling

inactive inference uh whenever you want to scale active inference to implement it to solve any kind of

problem in the world you want to do something because there's a lot of computations that are not

going to be tractable otherwise and so one um one cool example is when it's in the figure after this actually

so um when you want to for any kind of decision making inactive inference you need to approximate

the expected free energy and so uh no the one just before here uh yeah perfect um

um so if you look at the second equation there you get this minus log p of an action sequence

equals expectation of something this minus log p of the action sequence is our notation in this paper

for the expected free energy um so the expected i mean you probably all know this the expected free

energy is given by the expectation of something now um typically and when you have a very high dimensional

model which happens in in most applications i would say you get a very high dimensional expectation

what is an expectation an expectation is an integral with respect to a probability distribution

how do you compute expectations while you do that through sampling or at least um sampling is the way to

compute high dimensional expectations that's the most efficient in statistics and this is the reason why

um something is studied in statistics it's because that's how you solve these problems i mean

expectations just come up everywhere um just like so many things about machine learning

bow down to computing expectations and so here in particular the expectation that is very dear to

to us very close to our hearts is the one that gives you the expected free energy the expected free energy is the expectation of something so how do you compute that in a high dimensional model uh where you have to use something and so there comes uh the usefulness of hamiltonian monte carlo if you have a discrete state space model you're not going to be able to use hamiltonian monte carlo because hamiltonian monte carlo is on continuous state space models um it's on continuous state space right we talked

about continuous positions continuous velocity i talked about something a probability distribution on my desk which is a continuous state space so you're not going to be able to use hamiltonian monte carlo if you have a partially observed mark of decision process and you're doing active influence there but if you have for example i've been talking recently to ryan smith um who who does a lot of active inference and and he's really leading a lot of the modeling work with active inference and real data of patients of all sorts um many medical data and also using active inference to model psychological experiments and um one of the things he told me about well is okay well i have a bunch of data and partially observed mark of decision process is just not going to do it why is it not going to do it because the data that he has is lives on a continuous space i don't remember exactly what it was but let's just say for the sake of example that that data was you know the temperature in the room and and people had to the the subjects had to infer the temperature in the room based on their sensations so that's a continuous state space problem now imagine that you asked you asked someone to infer the temperature in a room then maybe you would change the temperature in the room or not and you would ask them again change the temperature or not and ask them again and so on in when you have this sort of setup you have a discrete uh you have a phenomenon that unfolds in discrete time because you just repeatedly ask some question and sometimes elapses between the questions um but you also have a state space that's continuous when you have this sort of data then the kind of model that you want to use is a partially observed mark of decision process that like we know and love but the state space is going to be continuous instead of this week the time is still going to be this week um now when you have this sort of model and i just want to say it as an aside or as a yeah as an aside that for the moment a lot of the modeling work and and simulation work in active influence uses discrete state space on the piece partially observed mark of decision process just because it's been sufficient and when you simulate agent is often like in great worlds and when you maybe have a state spaces in experiments they often by design discrete because it's just easier to handle but that's not something that that's not an assumption a simplifying assumption that we're gonna have that we're gonna be able to keep for you know that much longer i mean there's just so many things that you cannot account with these sort of models so one other model that is interesting to have is a partially observed mark of decision processes but with a continuous state space

so with those you get the problem of you know estimating the expected free energy which would be then an expectation or in other words an integration an integral in a continuous state space and so how do you do that efficiently if you are in a high dimensional state space where you use an Antonio Monte Carlo i think this would be really the the best method so this is how you join the dots

in this paper we didn't talk about sampling in discrete space because the methods are quite different so we had we really had to choose what to focus on there's of course a lot more methods in the literature than there are in this paper we just wanted to sort of review what were the main ones and what people really used in practice but you know when you have the expected free energy in a discrete state space computing the expectation that defines the expected free energy can often be a little bit easier because it comes down computationally to a bunch of matrix multiplications matrix vector multiplications which generally is doable unless i mean the state phase is enormously high and then we'd have to think of sampling methods in discrete space um but as soon as you move to a partially observed markov decision process in with a continuous space which i argue and ryan smith argues and i'm sure a lot of people have run into this um as soon as you have this kind of model well um yeah you just have an integration problem in continuous space and so Hamiltonian Monte Carlo is the way to go there awesome and your 2020 paper was a synthesis on some of the discrete state space formalisms of active so it's very interesting to see how you're now talking about where continuous time continuous state spaces can come into play and how interesting that active inference has uh the capacity to do deal with discrete and continuous state spaces and sometimes we lean on one leg or the other leg more but it spans the gap in a way that's actually like a value adding not like there's some sort of missing piece from one side or the other yeah yeah definitely i think active inference is great because it's so flexible in the sort of models that you can consider um but but yeah i mean historically the first models to be developed were continuous space and continuous time and then it works quite well because you can take gradients of the free energy and just minimize them over time um so you could you could get around basically everything by doing a gradient descent on free energy and then after that around like so that was like 2010 and then around 2015 uh people started thinking okay well how do i model um discrete time decision making with active inference and um and typically the decision making tasks at least the simple ones that are studied in neuroscience in behavioral neuroscience just to make things simple um the the number of actions that you have is discrete uh so you have a finite number of actions you have also and which is pretty small you would also have a finite number of things which would be pretty small and then and then people started to think okay well how do we use active inference to actually account for that um and so yeah and and so that's how the whole expected free energy and partially observed mark of decision processes uh came into play and now the community has grown a lot and there's more and more data that we want to account for there's more and more projects that are going on and with and you know people are realizing and i think i mean it's an obvious realization it's just not um it's not surprising at all that these two models developed in 2010 2015 they're just not going to account for everything and to think about other kinds of models and it really depends on what kind of data you have at hand and then one important and obvious type of model would be partially observed mark of decision process with

and actually um this is not a new thing i mean people have been using these kind of models um in reinforcement learning and control for a long time i don't know to what extent they've arrived at practically i mean they've sure arrived at practically implementable algorithms i don't know to what extent they're used to what extent their state of the art um but the point is they it's something that it's a kind of model that you know has existed for a long time um and so this sampling is would be a way to actually practically implement this and scale it when you want to put that within active inference so i think yeah it's an important thing to think about

[illegible]

representation in the figure of the running person and then how do the variables and the processes described in this figure relate to all of this you know five hours of talking about shadow hamiltonians and all of this right um yeah that's a broad question we might need another 10 hours you know to answer that in detail but uh just very short um well so the active inference section was meant to be as complete as possible even though it was very short and by complete by completeness i mean that we started by deriving active inference and deriving active inference from first principles um so this is really what the free energy principle does even though we weren't able of course to you know review the whole free energy principle in like one or two pages but we what we focused on was deriving the expected free energy from first principles and then from there you get the full active inference algorithm so proactive inference algorithm i think you just went past it's just below um if you go slightly below i think maybe page 29 or page 30 yeah uh still i think next yeah realizing adaptive agents yeah so this is the inference algorithm like 0.1 to 0.3 or something so just from a duration of the expected free energy um you actually get as a corollary from there the active inference algorithm that we know and love this is actually a more general version than what people use and um i'd be excited and i'm actually talking to people to actually implement that um but this is actually the most general version that has been established in the literature and it is more general in a meaningful way in the sense that all the beliefs all the probability distribution they are over trajectories or sequences of events so it's not only so so it means that all the computations they not only consider um events at a particular um time in the future for example but they consider trajectories so sequences of events and so this is a equivalent to considering different events at different points in time and their kind of dependencies and dependencies between them and so this is just to say that if you take if you would take then this algorithm and you would perform a mean field approximation over time which is saying that all the things that you see in the future are

sort of independent of each other at different points in time then you would recover what people typically use in the literature which is um easily implementable the the point i want to make here is that you can actually not have that limitation and have things that are a bit more um yeah a bit more complex that they are able to capture more complex relationships in the time series and input that they might be receiving like the kind of sensory data that they might be receiving and the kind of genetic models that

they would have um so anyway this is the active inference algorithm the most general one we derive that from first principles by deriving the expected free energy from first principles and so this relates us to to that figure that you just showed so that that figure that you showed um with s o and a is the starting point of the free energy principle described in this paper and so the point is that the world the point is that we want to describe decision making uh we want to describe actions as a function of sensations

and so and we want to come up with the most general description of actions as a function of sensations to be able to account for everything that's kind of the idea of the free energy principle

um so what do we do is we consider a world and a world in which there is an environment and an agent and so the environment here is denoted by s and s is a stochastic process why stochastic process because a stochastic process is the most general type of dynamic that exists at least as far as i know um it's really a random dynamic and it could be random in all sorts of ways it could also be non-random we sort of take the stochastic the random aspect as a extra ingredient just to include a lot more types of scenarios and things you can be confronted with so you partition the world into the agent which is o and a and the environment which is s and the agent then you subdivide it into two more components which are o and a

o are the what we call the observable state so these are like the sensations of the observation that you get at any point in time also a stochastic process and finally you have the autonomous states or you think about it loosely as the active states they turn out to be the active states in the implementation but so the active states they're really um you know like the muscles the things you can actually activate and make and then activate to you know influence the world um so we just partition the world into these three sets sets of states that interact and and evolve in some way they're both pretty stochastic processes that interact and then the goal of the free energy principle at least when applied to decision making is that you want to describe a as a function of o because what so what happens as an organism you can you have control over a you have access to o but you don't have the right control of o or your sensations and you don't have access to s because that's the environment and beyond beyond the mark of blanket beyond you beyond your envelope so you don't have access to s you know

o and you can control a a is what you can choose from and so the free energy principle just answers the question okay well um if i take this very general description of the world

how what is the equation of a as a function of o how can i describe a as a function of o

and so it turns out um there's some mild assumptions in this and these assumptions uh they're they're

guided by physical considerations about how humans are and how humans interact with the world but they're very mild

but whenever you take these assumptions you get that the active states or the autonomous states minimize expected free energy so so this is like a very succinct um derivation of the expected free energy from first principles

we start with the partition of the world we describe active states as a function of observations and so it turns out that the expected free energy is what describes the active states as a function of observations um and so and and so the basic active inference algorithm which everybody uses and which we have also in the paper is about it's all about computing the expected free energy and then selecting actions that minimizes expected free energy to the expected free energy as we saw i think it's in the next figure so the one with all the panels was within it a couple times already it's um it's an expectation with posterior distributions in it um what i mean by posterior distributions here is that that these are distributions conditioned on the history of the agent so on the observations that he has already seen and on the actions that he has already taken in any case um in any case so you have the expected free energy is an expectation with posterior distributions within so it means that to compute the expected free energy you have two problems you have computing posterior distributions through bayes and inference bayes rule and you then once you have them you can plug them in into this equation and then you need to complete the expectation and then you get the expected free energy so the first thing is about computing the posterior distributions now what has been proposed in the literature uh so far is well how do you

do you do approximate inference how do you actually approximate distributions you do that through variational

inference or approximate inference by minimizing free energy the the treatment here uh the derivation and we our aim was to do that provide a derivation that was as conceptually simple as possible

actually it highlights that the free energy is not the most important thing here

the important thing is really the expected free energy the free energy is just a tool to approximate the posterior distributions to then get the expected free energy but you could actually use any other type of divergence the free energy is just like a KL divergent plus some term that makes the whole thing tractable and so you can minimize the KL divergence to approximate the target distribution but actually what what this view highlights and by the way i'm not at all against the free energy i think it's really cool and whenever and there's so many methods to minimize free energy and if you can do that then fine and sure go for it like way to go but imagine you could not do that it would not at all be a problem because you could use any other kind of divergence uh to solve those inference problems um so so that would be the first step

you're going to approximate those posterior distributions by doing some kind of approximate inference which would be through minimization of the energy or through minimization of something else and then you have the second step which is computing the expectation now either you're in a low number of states discrete states from dp like thing and it's all a matter of vector matrix multiplications that are tractable whether you have a very very high number of discrete states and then you need to think about

sampling

in discrete states which we didn't discuss in this paper

either you are in a continuous state space and then you have an expectation in continuous state space

and then you want to think about an alternate Monte Carlo for example

so now with these two steps you have now an estimate of the expected free energy which gives you the quality

of any action sequence and so it's not only the quality but it's actually the negative log probability

of any action sequence regarding in this formalism because if you remember the one of the premise of all this was to describe actions as a function of sensations in a physical system in not even a physical system but in interacting stochastic processes

um and so the answer to that was well the expected free energy gives us you know gives us how actions relate to sensations

um the expected free energy and this is why we use the letter we use the we used minus log p of a as opposed to g to emphasize that the expected free energy is

not just like a function of action sequences but it is really the negative log probability of an action sequence

given some sensations um and so once you have computed the expected free energy you have this minus log probability of an action sequence

if you take the exponential negative of that the exponential negative of the expected free energy you get

um you get the probability distribution over action sequences um and so um by the way this exponential negative of expected

free energy is what we use all the time you might recognize this as the soft max of negative expected free energy

this is just all the time um kind of fundamental thing in active inference models um so i'm really talking about the same thing here

um so once you take the soft max of negative expected free energy you get p of a which is the distribution of reaction sequences and now you have two possibilities either you want to stimulate the most likely action sequence

in which case you want to simulate the the action sequence that maximizes the probability distribution

um so in effect you have an optimization problem you out of all um yeah you need to find the action sequence that is going to maximize this probability distribution that's given by the expected free energy or

or you want to simulate a typical action so what do i mean a typical action a typical action

action sequence is just a sample from the distribution and so if you want to sample from i mean if you want to do that

then you have a sampling problem where again you need to sample from from a distribution uh over action sequences given by the expected free energy so this is really how how all these methodologies connect um

um yeah i think i think that's kind of it yeah one last thing is um typically when we simulate active

inference we never do the sampling at the end we always take the action sequence that minimizes expected free energy in other words um in in other words the action sequence that maximizes that probability distribution

is the action sequence that um and this is because when we simulate things we it turns out i mean this is how people have done in the literature people are more interested in the most likely action sequence that um that an organism or an agent would produce but if you want to model data if you want to use active inference to model data then you actually want to not just simulate the most likely action sequence but simulate

any kind of action sequence that would fall out of this and so this is really the case where you need to sample from that final distribution as opposed to do optimization so depending on the use case you either have a sampling or an optimization problem after you've um you've computed the expected free energy so this is how these whole things um fall together and and so you might be asking okay well you talked about sampling and optimization there's also a section about inference where does that fit in and so it fits in to uh it fits in on on the remark i gave that you know we have these posterior distributions within the expected free energy how do we compute them how do we approximate them one way is by minimizing

free energy another way is by minimizing any other kind of divergence and in that section over there we just reviewed some kind of divergences that are very popular in the in the statistical inference literature mainly because they have desirable properties and and so if you weren't able um to use the variation of free energy for some reason or maybe something to be explored and something to be explored is just to use another type other types of divergences and and just see what happens um so the the open problem to be explored is whether we can get better performance by using different kinds of divergences and and see what we get uh maybe there's particle algorithms that are out there with these other types of divergences that we can make use of to do better performance to get better performance um can we actually describe and quantify the improvement in performance that we get by using these other types of algorithms when would these be appropriate and useful and these are all open questions um another open question that i think is interesting is can we model maladaptive behavior um by using different kinds of divergences that would not work as well as the free energy and there's um there's an interesting paper on that by uh newer sejid i think and colleagues i think it's called um basin brain brain basic brains and the rainy divergence it's been published on neural computation i think last year or or the year before and so in that paper instead of using the variational free energy to approximate all these posterior distributions um she used uh the rainy divergence which is which generalizes the kale divergence in some way and show that for different rainy divergences you get different types of approximate posteriors and she basically um looked into yeah what kind of differences you get from there in terms of perception in terms of decision making and she showed that for that particular divert divergence i think the conclusion was that you basically get different phenomenology you get different behavior uh so just something to be explored i think the option was that maybe i don't remember how far the paper went in this but i think kind of the the goal was to model maladaptive behavior using some kinds of rainy divergences that did not work as well as the free energy so this is to say so this is an interesting work um one could examine other kinds of divergences and see whether you actually get better or worse performance than with the free energy uh one thing one word of

caution though is this the free energy is just so good in the sense that it uses the KL divergence and as I mentioned at the beginning the killer has so many properties and it's just so fundamental so um it's not straightforward to see that when you first come up with it but I guess the more I read in different disciplines and the more the KL divergence comes up and the more the more I see properties of KL divergence in different disciplines that just make it so interesting like for example the KL divergence in statistical physics is the relative entropy so it quantifies the amount of entropy that a distribution has with respect to another entropy as we know is just a very fundamental thing in information theory the KL divergence quantifies the difference in information between two distributions the amount of bits that you would need to I mean um if you take two distributions a and b uh the KL between the two between the two quantifies the amount of information uh that it takes to go from one to the other so now you might object and say okay but $KL(a, b)$ is different from $KL(b, a)$ and so how can it be that you know it quantifies the amount of information um they need to go from a to b or b to a because it's not symmetric um so the answer is it's either $KL(a, b)$ or $KL(b, a)$ that has this meaning or $KL(b, a)$ that has this meaning and I just can never remember which direction but it's one of them that has this uh interpretation in terms of uh the difference in information um anyway um so KL is just like um something it just comes up everywhere and it's just um so useful and it has all of these nice properties which makes the free energy really um a construct of choice um but that said there's other divergences one of them which we describe a bit in the paper called um the maximum mean discrepancy which I think also has nice properties it's not necessarily well it's a bit different it's very different in terms of how you construct it from the KL divergence but I think my current understanding right now and um my current understanding right now is that if you cannot use scale then maximum mean discrepancy is just like really nice um but but yeah so the word of caution was that probably most divergences out there are not going to do as good of a job as a free energy uh but some clever ones might and it's an open problem of you know determining which and and yeah so this is how you know we get the link between all these different sections so to to do perception we need inference to do decision making that is computing the expected free energy we need something typically and then to do action selection we either need sampling or optimization and also inference is uh an optimization of beliefs sampling also as we've discussed is also an optimization of probability distributions you can see that as an optimization of beliefs as well so it's all very tightly interconnected wow thank you lance that's very informative brings up a lot of ways to go and it really shines a different light on even what learning active inference or learning free energy principle it was a roller coaster just listening when you describe the figure of the running person and the new generalized representation here which is so sparse it looks like it's pseudocode but it's actually basically necessary and sufficient to describe it is it's pseudocode yeah it is pseudocode and it is also exactly like this is really what active

inference boils down to and that was kind of like where we got at we're like okay well we have this mess of papers of active inference not that they're actually a mess but you know there's so much information and we're you know really thinking reading all this the latest free energy principle literature that i've also worked on a lot and also the active inference literature and really thinking okay well how do we strip all that of the neuroscience how do we strip all that of the cognitive science how do we just retain the math and present it in the simplest way possible that would be appealing for a mathematician and also hopefully for a computer scientist it turns out that we got a way simpler perspective than anything that's anything that's out there i think um and really yeah so this is the active inference algorithm in pseudocode and in detail also in in a way so i totally agree again thanks for the work and for sharing it this way because it is the fewest pixels for the highest resolution picture so so then to describe the fundamental cybernetic challenge as a partition an axiomatic partition which one can then say also has grounding in the spatial temporal boundaries of the world or in geometric boundaries in informational spaces but the particular partition is used to separate the map not necessarily making claims about the territory and the actual nature of the objects and their articulations or anatomy but on the map which we get to construct we make it in a way that's amenable to the particular partition which is not too much more than the separation of figure from ground or agent-based modeling it's just a separation of some autonomous entity from some external process then the task of the free energy principle applied to decision making as you said was to describe action as a function of observation and everything in between is broadly considered cognitive but for any given system it's going to play out in this immensely nuanced way with a lot of bespoke mechanisms and why do we

take that particular partition step well it's kind of like read write access in a computer system there's things we don't have access to hidden states and also the reverse of that which is we can't access hidden states nor them influence or affect us so i think of that as like no telekinesis no telepathy you can't go directly across the blanket you have to intermediate through the blanket of observation action and then with respect to our particular states our blanket and internal states we have access to observations but not direct control nor necessarily would we want to because if we had the lever to change what we directly received then algorithms might learn strategies that basically self-deceive so that the observations look good look good look good until the whole system crashes so we want access to observations but not direct control and then the autonomous states are what we in the optimal ideal situation have total control of which is like our mind and our body what we think and what we do it turns out through the pragmatic turn and the inactivist insights in cognitive science that a lot of action sequences have to do with changing what observations are sought after like epistemic affordances so there is an enmeshment of action but it's also really important that we have access but not control of observations we don't want to have like our control on the thermometer but we want the best possible thermometer and then we want to control the best possible interpretation that's like signal processing and the best possible action sequence which is like decision making and control theory and then free energy principle is addressing that question what is the equation of action as a function of

observation but then it was quite a roller coaster when you said that the free energy wasn't even necessarily the only way to do it but it's absolutely true it's the properties of the free energy are inherited from one of the terms being the Kullback-Leibler divergence any other having the ability to basically ignore in certain relative expected free energy calculation contexts so then in that situation differences in free energy do come down to differences in the Kullback-Leibler which does have all these properties but that doesn't mean that the free energy is itself axiomatically posited it's actually downstream of the particular partition to use free energy or anything like that at all and other discrepancies may have other properties in different ways exactly and and so there's a bit of a nuance in the sense that here we presented the stimulus version of the free energy principle just describing decision making as you as you summarize so describing a as a function of o but if we and and by the way about that so we're we didn't even talk about Markov blanket in this paper um i would i would say that the Markov blanket is under the hood because we're saying okay well these are the states that you have access to and these are the states that you do not the states that you have access to are a and o the states of the agent the states that you do not have access to are the environment so in some sense there is a Markov blanket but we didn't even mention we didn't even have to mention that in the paper it's just you partition the world into three sets of states s o and a um that are by definition a um and you just yeah just from this tripartition you want to describe one as a function of another ignoring the third um and so that's yeah that's how you derive expected free energy you see that um action sequences are described by the expected free energy and these this is a function of sensations um if we went further in the and so this is just what you need crucially this is just what you need to derive the active inference algorithm if you if we went further in reviewing the free energy principle then we would see the fundamental role that the free energy plays um and so actually when you know reading the latest papers on the mathematical theory of the free energy principle the role of the variation of free energy is pretty clear um but here the point is that if we just care about decision making if we just care about the normal active inference algorithm then we don't even need the variation of free energy so sure uh the free energy should be preferred why not uh if it is available but if it is not for some reason then there's no reason why not to use another kind of the kind of the evidence so very interesting it's making me think about linear regression and the sum of squares the L_2 norm is one approach that's often used to fit a regression line because it has good optimization properties there's good software packages there's good education there's good communication around it and so on but one can select other norms and choose to fit a linear regression with an L_1 norm or with an L_3 norm and so that entire question of fitting the linear regression is a degree of freedom how the regression is fit that's downstream of a commitment to for example model a system in a generalized linear modeling framework and so analogously the upstream commitment or the first principles which yes can be understood as axiomatic and also have some empirical status in terms of this partitioning a figure from ground the particular partition can simply be accepted axiomatically which is to say without appeal to evidence

or somebody might have another upstream axiom and choose to model uh things according to a particular partition

from there just like we could have chosen the l123 norm there are different discrepancy criteria data or measures that we might want to use and some of them apply better or worse or not at all depending on what software hardware data set and generative model we have and so it makes sense to that pull pull pull back into the the upstream understanding of active inference as a process theory for particular partitioned systems and then for those who want to engage in the modeling to have that uh discussion about the garden of the branching paths well you could use a discrete timer you could use a continuous time and then from here you could do this sampling or you could do that one and if we have this computer access we can do that but if we have to do it this way we'll do it like that and that's all operational and logistical but it's actually all under the umbrella or under the auspice of the theoretical or conceptual commitments that are actually not being questioned once one is in that modeling discussion just like you could have the l123 norm conversation and maybe a reviewer asks you why you chose the two norm versus the three but it's a broader level of questioning why one took on the linear modeling framework at all and our analogous upstream bottleneck not in terms of rate limiting but just in terms of like eye of the needle is the particular partition

yeah definitely and i just want to add something actually about the choice of divergences uh or the choice of discrepancy that you might want to use to solve the inference problem i think the analogy would linear regression is a really good one um when you do linear regression yeah you can use all types of norms and it's really a design choice and here in the algorithm you also have that design choice are you going to use kl so free energy to do these inferences uh by the way the inferences are really uh equation 43 and 44 which are approximate um actually you had them in the slide already oh 43 and 44 oh oh here got it yeah yeah um which are approximate um you know some posterior distribution with an approximate posterior distribution so you can do that with the free energy which is the same as the kl divergence or you could do that with a whole bunch of other divergences one that i mentioned and that i think is particularly interesting is the maximum mean discrepancy and so um here's the difference between kl and maximum mean discrepancy at least like i guess a important conceptual difference so the kl when you look at distributions that are very close to each other it's going to measure it's basically going to become symmetric when the distributions are very close and it's going to measure the amount of information that differs between them the maximum mean discrepancy when you take two distributions that are

very close it's going to it it reduces to what people call the earth movers distance also called like the vast time distance so here is the intuition if you take two distributions that are very close just imagine distribution a as a pack of dirt and distribution b as a pack of dirt or or a stand with some shape uh the maximum mean discrepancy is gonna tell you the amount of work that you need to put all that dirt

from distribution a and pile it in the shape of distribution b so of course there's so many different ways in which you could take all that dirt from distribution a and remodel it in distribution b but you're interested in

the

minimal um the minimal energetic cost that would take so like the optimal way of doing that movement people call that optimal transport um now if you if you're familiar with optimal transport you know that the the vast time distance um is a distance between probability distributions that regardless of how far they are is going to measure um like the it is going to give you this um cost of optimal transport so the energetic cost of moving all that um you know pack of dirt from place a to place b in the optimal way um so the vast time distance is something that we could use here um but maximum mean discrepancy it has a lot of very nice properties and basically reduces to the vast time distance when we consider very close distributions so uh this is not something that just a mathematical curiosity but it says that when one builds when one builds a distance out of a divergence what one does is one takes you know very close distributions measures the divergence between them at which point the divergence is pretty much symmetrical

and then you basically keep adding divergences along a trajectory until you get to the final one and this way by aggregating divergences between very close distributions and doing that and you know adding

that along a trajectory you get a distance a meaningful distance between you know distributions that could be very far if you do that with the KL divergence you get what's called the fisher information distance which measures really the um really um yeah the the amount of information that took you from go to go from one distribution to another distribution regardless of how far they are

um if you do that with advanced search time distance you will get the optimal transport cost so the amount of energy that you need to put all that dirt they say to place me in the optimal way if you do that with the maximum discrepancy you would also get that you will also get the optimal transport thing so um i just so the yeah i just want to say that the maximum discrepancy meaning discrepancy between two distributions that are far away will not coincide with the Wasserstein distance which gives you this optimal transport cost but when you actually derive a distance from these divergences the distance derived from the maximum meaning discrepancy and the Wasserstein distance will end up being the same and um this distance that you actually get or the topology that is derived from the distance so by topology a topology is really a notion of closeness um and so to understand closeness you need to understand you know infinitesimal distances um so the infinitesimal distance is derived from Wasserstein or MMD they're the same

the topology that results is the topology of what people call weak convergence which is the standard topology that's considered in probability theory um so MMD and Wasserstein they are very natural in that sense

in the sense that they they they uh they people say that they met rise with conversions like they give give rise to the standard topology between probability distributions um so coming back to the choice of divergences if you're interested in approximating distributions in the sense of approximating their information content then KL is very natural but if you were for some reason and maybe would not be in this kind of application but another kind of application if for some reason you're interested in approximating

distributions for the sake of how close they are when you look at them uh then then you would use MMD or or best of shine wow great information and i'm just thinking about we're switching lanes on the highway we're trying to get from here to there yes we want to know about the informational closeness of this tale of two densities but we also want to know about the transport closeness because we have a schedule and a budget and decisions to make and there are trade-offs so being able to move the earth optimally while we're switching lanes and accelerating and slowing down i know this is mixing many angles informally we want to have a lot of options for how to think about that challenge of moving dirt between the tail of two densities and make sure everybody is driving in the right lane at the right time

definitely and yeah it's a it's a design choice and it's an important one because it depends on you know what kind of properties we want to preserve and um yeah so all these all these divergences some some are not so useful i guess but um some of them they're they're just very natural and the westerstein and mmd they're very natural in that sense um so yeah very important to keep in mind um and not only for active inference uh but just you know in general for any kind of inference problem

well let us each have a closing round or reflections any thoughts or any next steps or any suggestions or other information you'd like to provide um it's hard to say i feel we've you know we've covered so much already um there's still i think to me what's most exciting about is um scaling active inference right now um because you know you get this active inference algorithm in the paper uh that's derived from first principles and just from the derivation you see okay well there's actually so many things i mean the assumptions are so small that there are so many things that you can model with this active inference algorithm and basically you will oh and all the heavy lifting is done by the genitive model so if you use one derivative model you will get one behavior if you use another genitive model you will get another behavior but all the equations will remain the same um so so it speaks to generality now in in having something that's very general you get something that's also very non-specific and i guess for for neuroscientists and people who are interested in intelligence you just um generality comes at a cost of being you know non-specific and non-specific about the brain in general um so really the big question to me uh long term is what kind of genitive models do we need to simulate brain-like behavior because really this is really the interesting behavior or the most interesting behavior um so it speaks to a big research program that a lot of people are carrying and have been carrying for a long time but you know um it's about what kind of representations do we have of the world what kind of priors do we have what kind of uh can we identify the priors that we are born with there's a lot of research on um computational or just playing cognitive science like normal cognitive science just studying babies and seeing what kind of priors what kind of you know basic information they have when they come out of the womb there's a lot of things that they can already do there our genetic code is preconditioning us to operate efficiently in this world and be able to flexibly adapt to any kind of situations that might arise in the natural world and so it's a huge research program to you know understand and model these priors and um yeah and not only the priors but also the likelihood all the representation all the state spaces so i think that's the way forward the green energy principle is very elegant

Thank you.